

From Tools to Teams: A Hands-on Lab on LLM Agent Protocols

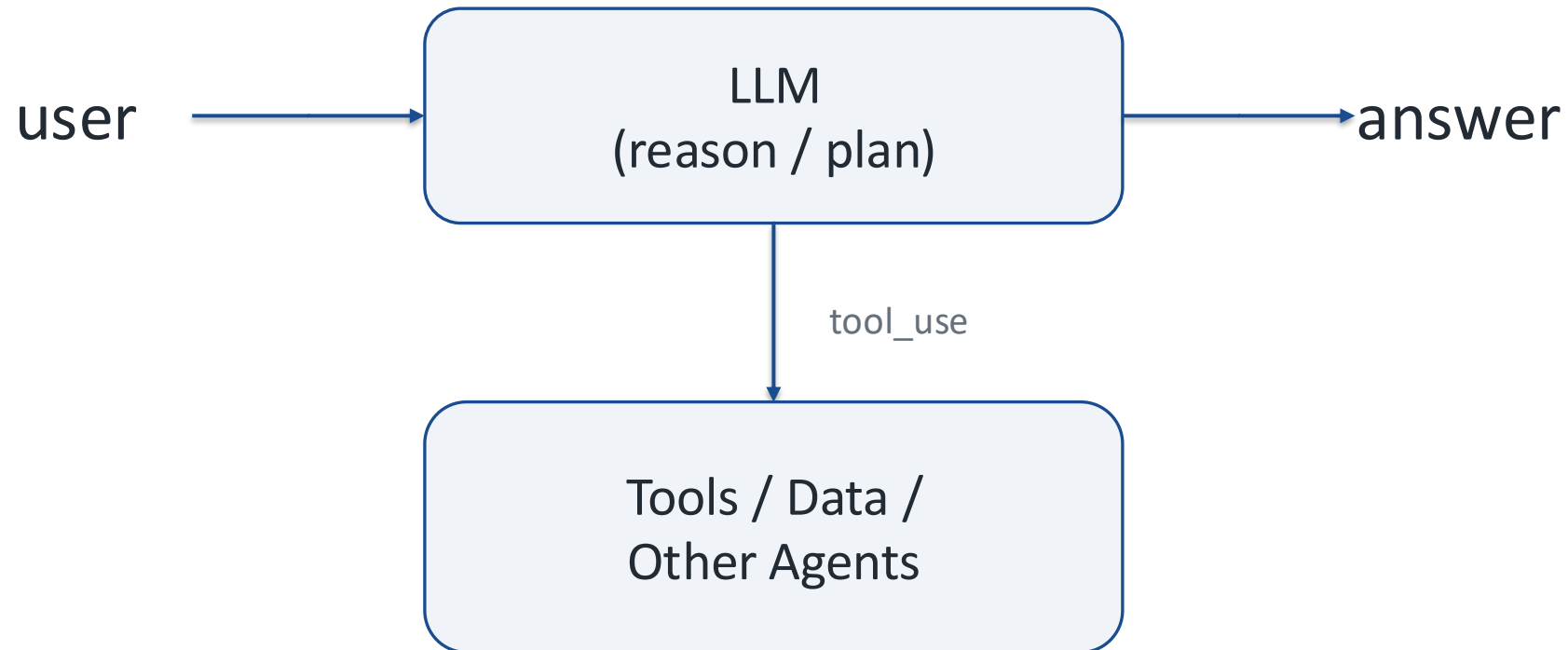
MCP · A2A · FA

Dekun Wu

PhD student in Computer Science (Supervisor: Prof. Bang Liu)

Université de Montréal & Mila

What is an LLM agent?



LLM + tools + state + an autonomous decision loop.

Three protocols, three scopes

Scope grows →

MCP

Model Context Protocol
by Anthropic

LLM ↔ tools / data

- Agent calls fixed functions
- Schemas auto-generated
- Stable, mature ecosystem

A2A

Agent-to-Agent
by Google

Agent ↔ Agent

- Callee is itself a smart agent
- Self-description via Agent Card
- Tasks have a lifecycle

FA

Foundation Agents Protocol
by DeepWisdom

Humans / Agents / Tools / Services

- Federated network of entities
- Discovery + persistent identity
- Trade primitives built in

MCP — agent to tools



All data routes through the LLM. Servers don't talk to each other.

A server exposes

- **Tools** — callable functions
- **Resources** — readable data

Use it when ...

- Reach a DB, script, or internal/external API
- Reuse one tool across many LLM clients
- Want schemas from code, not hand-written

MCP is the calculator's keypad.

MCP — schemas auto-generated

Function signature + docstring → JSON schema

What you write (Python)

```
from mcp.server.fastmcp import FastMCP
import yfinance as yf

mcp = FastMCP("stocks")

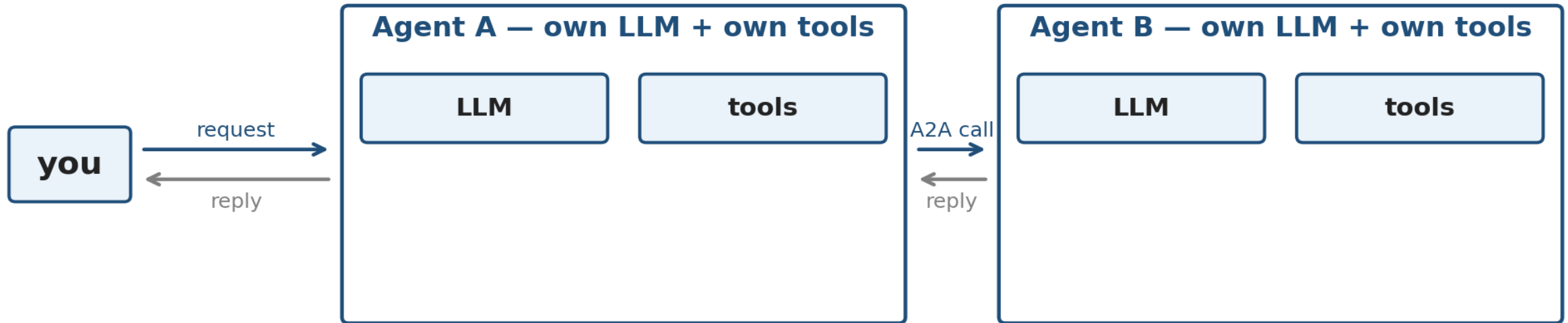
@mcp.tool()
def get_quote(ticker: str) -> dict:
    """Get the current stock quote for a ticker."""
    return yf.Ticker(ticker).info
```

What the LLM sees (JSON)

```
{
  "name": "get_quote",
  "description":
    "Get the current stock quote for a ticker.",
  "inputSchema": {
    "type": "object",
    "properties": {
      "ticker": { "type": "string" }
    },
    "required": ["ticker"]
  }
}
```

One decorator. No hand-written schemas

A2A — agent to agent



Three core concepts

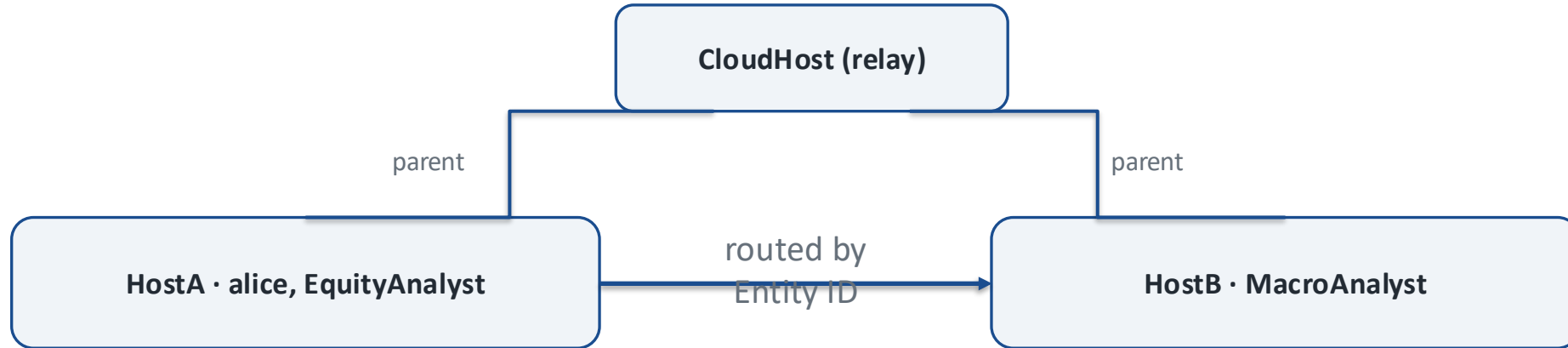
- **Agent Card** — self-description at a URL
- **Task** — work unit with a lifecycle
- **Message / Artifact** — input / output

Use it when ...

- Callee should plan on its own
- Long-running, async work
- Caller & callee owned by different teams

A2A is a phone call to your friend.

FA — humans, agents, tools, services



What FA adds

- **Entity** — HUMAN / AGENT / TOOL / ORG
- **Host** — can be pure routing
- **Friend** — persistent relationship
- **Trade** — contracts, payments, ratings

Use it when ...

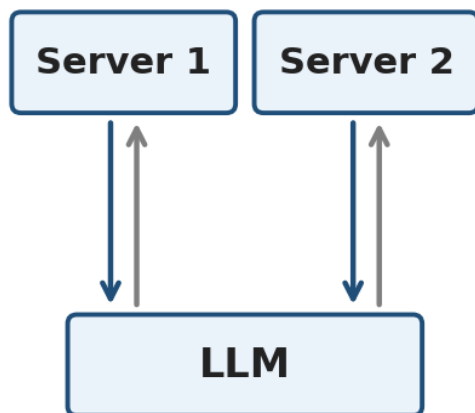
- Cross-org / cross-deployment
- Need offline delivery + central audit
- Identity must survive machine changes

FA is the email system for agents.

Side by side — three architectures

MCP

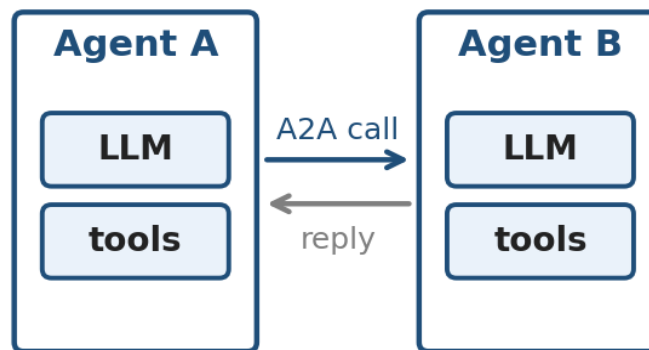
LLM ↔ tools



LLM is the only initiator.
Servers only respond.

A2A

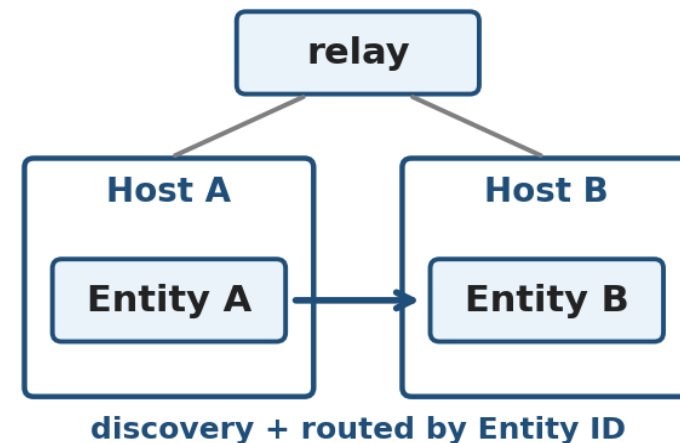
Agent ↔ Agent



Callee is its own agent.
URL must be pre-known.

FA

Network of entities



Address by Entity ID.
Discovery in protocol.

MCP vs A2A vs FA — at a glance

	MCP	A2A	FA
Connects	LLM ↔ tools	agent ↔ agent	network of entities
Identity	tool name	URL + AgentCard	stable Entity ID
Discovery	static `tools/list`	out-of-band URL	protocol primitive
Long-running	sync calls	Task lifecycle	Mail; build on top
Offline	—	call fails	mailbox + auto-flush
Reachability	local stdio / HTTP	needs inbound	outbound WebSocket
Maturity	mature SDK	2024, growing	research

Lab agenda

#	Topic	Time	Notebook
1	MCP — Claude with stock-data + plotting tools	~10 min	demo1_mcp/01_mcp_demo.ipynb
2	A2A — delegate "Analyze NVDA" to an equity analyst agent and a macro analyst agent	~10 min	demo2_a2a/02_a2a_demo.ipynb
3	FA — equity analyst discovers + calls a macro analyst on another host	~10 min	demo3_fa/03_fa_demo.ipynb

Same input across all three: a question about NVDA.

Let's begin →

demo1_mcp/01_mcp_demo.ipynb

demo2_a2a/02_a2a_demo.ipynb

demo3_fa/03_fa_demo.ipynb

References

MCP: <https://github.com/modelcontextprotocol>

A2A: <https://github.com/a2aproject/A2A>

FA: <https://github.com/FoundationAgents>

Q&A

Repo: <https://github.com/jackwu502/ivado-protocol>

Contact: dekun.wu@umontreal.ca



Thank you.